

2023 IEEE International Conference on Blockchain (Blockchain) **Blockchain 2023**

Table of Contents

Message from the Blockchain 2023 Steering Chairs	xiv
Message from the Blockchain 2023 General Chairs	xvi
Message from the Blockchain 2023 Program Chairs	xviii
Organizing Committee	xx
Program Committee	xxi
Sponsors	xxv

Short Papers

Blockchain-1: Security, Privacy, and Trust (I)

Incentive Attacks on DAG-Based Blockchains with Random Transaction Selection	1
<i>Martin Perešini (Brno University of Technology), Federico Matteo Bencic (University of Zagreb), Martin Hruby (Brno University of Technology), Kamil Malinka (Brno University of Technology), and Ivan Homoliak (Brno University of Technology)</i>	
Understanding the Cryptocurrency Free Giveaway Scam Disseminated on Twitter Lists	9
<i>Kai Li (San Diego State University), Darren Lee (San Diego State University), and Shixuan Guan (San Diego State University)</i>	
The Validator's Dilemma in PoW Blockchain: An Evolutionary Game Perspective	17
<i>Liang Liu (University of Tsukuba, Japan), Qiao Wang (University of Tsukuba, Japan), Stephen John Turnbull (University of Tsukuba, Japan), and Kazumasa Omote (University of Tsukuba, Japan)</i>	
Mitigating Undercutting Attacks: Fee-Redistribution Smart Contracts for Transaction-Fee-Based Regime of Blockchains with the Longest Chain Rule	25
<i>Rastislav Budinsky (Brno University of Technology), Ivana Stancikova (Brno University of Technology), and Ivan Homoliak (Brno University of Technology)</i>	
A Benchmark for Different Implementations of Zero-Knowledge Proof Systems	33
<i>Max Kobelt (Hamburg University of Technology, Germany), Michael Sober (Hamburg University of Technology, Germany), and Stefan Schulte (Hamburg University of Technology, Germany)</i>	

Blockchain-2: Security, Privacy, and Trust (II)

The Blockchain Trilemma Described by a Formula	41
<i>Taishi Nakai (Kyoto University, Japan), Akira Sakurai (Tokyo Institute of Technology, Japan), Shiori Hironaka (Kyoto University, Japan), and Kazuyuki Shudo (Kyoto University, Japan)</i>	
Statically Checking Missing Input Validations in Solidity Smart Contracts - A Case Study	47
<i>Sundas Munir (Halmstad University, Sweden), Mirza Sanam Iqbal Baig (Eagle Games Sweden AB, Sweden), Mah Noor (Halmstad University, Sweden), and Syeda Hina Murad (Halmstad University, Sweden)</i>	
Multi-blockchain Model for Data Sharing with Bell-LaPadula Access Control	55
<i>Hsiau Chuen Chong (Macao Polytechnic University, China) and K. L. Eddie Law (Macao Polytechnic University, China)</i>	
ALGOMOVE - A Move Embedding for Algorand	62
<i>Lorenzo Benetollo (Ca' Foscari University of Venice, Italy; University of Camerino, Italy), Michele Bugliesi (Ca' Foscari University of Venice, Italy), Silvia Crafa (University of Padua, Italy), Sabina Rossi (Ca' Foscari University of Venice, Italy), and Alwise Spanò (Ca' Foscari University of Venice, Italy)</i>	

Blockchain-3: Applications and Services (I)

Blockchain-Based Identity Authentication Oriented to Multi-cluster UAV Networking	68
<i>Zesong Dong (Xidian University, China), Wei Tong (Xidian University, China), Zhiwei Zhang (Xidian University, China), Jian Li (Xidian University, China), Weidong Yang (Xidian University, China), and Yulong Shen (Xidian University, China)</i>	
Pricing Mechanism of Non-Fungible Token (NFT) Driven by Rarity Design	74
<i>Wenze Xiong (Xi'an Jiaotong-Liverpool University, China), Yetong Wang (Xi'an Jiaotong-Liverpool University, China), Wanxin Li (Xi'an Jiaotong-Liverpool University, China), Jie Zhang (Xi'an Jiaotong-Liverpool University, China), and Hao Guo (Northwestern Polytechnical University, China)</i>	
Joint Pricing for Carbon and Electricity Trading in Blockchain-Based Smart Grids: A Two-Stage Game Approach	80
<i>Wang Xu (Zhejiang Normal University, P. R. China), Feilong Lin (Zhejiang Normal University, P. R. China), Riheng Jia (Zhejiang Normal University, P. R. China), Changbing Tang (Zhejiang Normal University, P. R. China), and Minglu Li (Zhejiang Normal University, P. R. China)</i>	
Mitigate Gender Bias in Construction: Fusion of Deep Reinforcement Learning-Based Contract Theory and Blockchain	86
<i>Zijun Zhan (China University of Petroleum (East China), China), Yaxian Dong (The Pennsylvania State University, USA), Daniel Mawunyo Doe (University of Houston, USA), Yuqing Hu (The Pennsylvania State University, USA), Shuai Li (University of Tennessee, USA), Shaohua Cao (China University of Petroleum (East China), China), Wei Li (Texas Southern University, USA), and Zhu Han (University of Houston, USA)</i>	

AgBIS: A Blockchain-Enabled Crop Insurance Platform Against Adverse Selection, Moral Hazard, and Insurance Frauds	92
<i>Zhonghao Liao (Iowa State University), Chaoqun Lu (Iowa State University), Mark Mba Wright (Iowa State University), Hongli Feng (Iowa State University), and Yong Guan (Iowa State University)</i>	

Workshop Papers

Blockchain-4: BSS Workshop (I)

An Evaluation Framework for Assessing IPFS Performance Within a Blockchain-Based Healthcare System	100
<i>Ghassan Al-Sumaidae (ECE - McGill University, Canada), Rami Alkhudary (Universit�e Paris-Panth�eon-Assas, LARGEPA, France), Zeljko Zilic (ECE - McGill University, Canada), and Andraws Swidan (Department of Computer Engineering, University of Jordan, Jordan)</i>	
A Video Security Verification Method Based on Blockchain	105
<i>Zhenghang Zhao (Zhengzhou University), Yunxia Liu (Zhengzhou Normal University), Hongguo Zhao (Zhengzhou Normal University), and Yonghao Wang (Birmingham City University)</i>	
A Proxy-Layer Approach to Secure Smart Contract Deployment on Private EVM-Based PoA Blockchains	109
<i>Yonghao Wang (Birmingham City University, UK), Jahid Ali (MEER Labs Limited, UK), Junaid Arshad (Birmingham City University, UK), and Yunxia Liu (Zhengzhou Normal University, China)</i>	
Omniscient: The Universal Blockchain Oracle	113
<i>Kristi�n Ko�t�l (Slovak University of Technology in Bratislava, Slovakia), Igor Durica (Slovak University of Technology in Bratislava, Slovakia), and Michal Ries (Slovak University of Technology in Bratislava, Slovakia)</i>	

Blockchain-5: BSS Workshop (II)

MACS: A Multi-asset Coin Selection Algorithm for UTXO-Based Blockchains	121
<i>Gholamreza Ramezan (Cardano Foundation, Switzerland), Manvir Schneider (Cardano Foundation, Switzerland), and Mel McCann (Cardano Foundation, Switzerland)</i>	
Trustworthy VANET: Hierarchical DAG-Based Blockchain Solution With Proof of Reputation Consensus Algorithm	127
<i>Zhongxu Dong (University of Glasgow, U.K.), Huanyu Wu (University of Glasgow, U.K.), Zongyao Li (University of Glasgow, U.K.), De Mi (Birmingham City University, U.K.), Olaoluwa Popoola (University of Glasgow, U.K.), and Lei Zhang (University of Glasgow, U.K.)</i>	
Blockchain International Certification Layer	133
<i>Saeed Alketbi (Universiti Utara, Malaysia; Dubai Customs, UAE) and Massudi bin Mahmuddin (Universiti Utara Malaysia, Malaysia)</i>	

Blockchain-6: FBS + SPB Workshop

Demo Paper: Anonymous Authentication on Trust in Blockchain-Based Mobile Crowdsourcing System	140
<i>Guocang Liu (Xidian University, China), Haoxiang Han (Xidian University, China), Wenxiu Ding (Xidian University, China), Shufan Fei (Xidian University, China), and Zheng Yan (Xidian University, China)</i>	
EMIS: A Uniform Management and Resolution System of Multiple Identifiers on Ethereum	145
<i>Feng Zhao (Peking University, China), Hui Li (Peking University, China), Shaoliang Peng (Hunan University, China), Xiyu Wang (ZTE Corporation, China), Ping Lu (ZTE Corporation, China), Qianbin Chen (Chongqing University of Posts and Telecommunications, China), Weixiang Huang (China Mobile Internet Co., Ltd., China), Jianping Wu (China Communication Service Construction Co., Ltd., China), Xiang Zhu (China United Network Communications Group Co., Ltd., China), Selwyn Deng (China Unicom Global Limited, China), Hanxu Hou (Dongguan University of Technology, China), Jieren Chen (Hainan University, China), and Han Wang (Peking University, China)</i>	
Unveiling Vulnerabilities in DAO: A Comprehensive Security Analysis and Protective Framework	151
<i>Chia Cheng Tsai (National Taiwan University), Cheng Chieh Lin (National Cheng Kung University), and Shih Wei Liao (National Taiwan University)</i>	
SrNFT: Simple Restricted Non-Fungible Token Framework for Transactions with Risk Detection... ..	159
<i>Zixuan Liu (Hainan University, China), Yirui Bai (Hainan University, China), Tiankai Xu (Hainan University, China), Haoyu Gao (Hainan University, China; Oxford-Hainan Blockchain Research Institute, China), and Hong Lei (Hainan University, China; SSC Holding Company Ltd., China)</i>	
Efficient Authentication System Based On Blockchain Using eID Card	166
<i>Liang Liu (University of Tsukuba, Japan) and Kazumasa Omote (University of Tsukuba, Japan)</i>	
A Cross-Chain Identity Authentication Scheme Based on DID	172
<i>Yirui Bai (Hainan University, China), Zixuan Liu (Hainan University, China), Xuefeng Liu (Xidian University, China), Xiang Lu (Chinese Academy of Sciences, China), and Hong Lei (Hainan University, China; SSC Holding Company Ltd., China)</i>	

Blockchain-7: IBTA Workshop (I)

Data Interoperability Technology Based on Chain-Network Convergence	180
<i>Haoyu Wang (University Beijing, China), Jiping Xu (Chinese National Light Industry, China), Xin Zhang (Chinese National Light Industry, China), Cheng Chi (Institute of Industrial Internet and Internet of Things, China Academy of information and communication, China), Zihang Yin (Institute of Industrial Internet and Internet of Things, China Academy of information and communication, China), and Yanjie Shao (Institute of Software, Chinese Academy of Sciences, China)</i>	

A Trusted Sharing Model for Risk Information of Food Full-Process and All-Information Based on Blockchain and Federated Learning	186
<i>Xin Zhang (Beijing Technology and Business University, China), Xueze Tan (Beijing Technology and Business University, China), Jiping Xu (Beijing Technology and Business University, China), Song Luo (China Academy of Information and Communications Technology, China), and Zhibo Qi (China Academy of Information and Communications Technology, China)</i>	
Trusted Access Control Mechanism for Intelligent Manufacturing Based on Decentralized Identifier	192
<i>Ru Huo (Beijing University of Technology, China), Dong Ni (Purple Mountain Laboratories, China), and Zihao Shao (Purple Mountain Laboratories, China)</i>	
The Appliance of Decentralized Identifiers in Zero Trust Network	198
<i>Tongtong Cheng (China Academy of Information and Communications Technology, China), Cheng Chi (China Academy of Information and Communications Technology, China), Yuwen Zhang (China Academy of Information and Communications Technology, China), and Zihang Yin (China Academy of Information and Communications Technology, China)</i>	

Blockchain-8: IBTA Workshop (II)

A Blockchain Based Trustworthy Space in Cross-Domain Synergy	203
<i>Lin Sun (Unicom Digital Tech., China), Xiaoyun Jia (Unicom Digital Tech., China), Jiaxi Liu (Beijing University of Posts and Telecommunications, China), Yulun Song (Unicom Digital Tech., China), Yunlong Xie (Unicom Digital Tech., China), and Chaoyue Xue (China Unicom, China)</i>	
Research on the Development of Blockchain-Based Distributed Intelligent Healthcare Industry: A Policy Analysis Perspective	209
<i>Yang Yue (Institute of Science Studies and S&T Management, Dalian University of Technology, China) and Joseph Z. Shyu (Institute of Science Studies and S&T Management, Dalian University of Technology, China)</i>	
The Ecological System of Digital Asset Markets: Based on the Perspective of Asset Trading and Valuation	215
<i>Cheng Chi (China Academy of Information and Communications Technology, China), Wenqu Chen (China Academy of Information and Communications Technology, China), Hang Liu (Central University of Finance and Economics, China), Xiaozhuang Li (Central University of Finance and Economics, China), and Fuyu Meng (Central University of Finance and Economics, China)</i>	

Full Papers

Blockchain-9: Applications and Services (II)

Blockchain-Based Trustless Fair Payment Protocol for Verifiable Confidential Outsourcing Computation	221
<i>Hiu Long Lee (The University of Hong Kong, China; The University of Hong Kong Shenzhen Institute of Research and Innovation, China), Man Ho Allen Au (The Hong Kong Polytechnic University, China), and Shi-Feng Sun (Shanghai Jiao Tong University, China)</i>	
Securing IoT Firmware Dispatch Systems with Blockchain	229
<i>Vince Biró (Technical University of Denmark, Denmark), Wei-Yang Chiu (Technical University of Denmark, Denmark), and Weizhi Meng (Technical University of Denmark, Denmark)</i>	
iBCTrans: A Practical Blockchain-Based Framework for Cellular Vehicular-to-Everything Networks	239
<i>Wei Yao (New Jersey Institute of Technology, USA), Yuhong Liu (Santa Clara University, USA), Fadi Deek (New Jersey Institute of Technology, USA), and Guiling Wang (New Jersey Institute of Technology, USA)</i>	
Blockchain-Based Smart Contracts for Land Title Registry Opportunities and Adaption for Fiji	247
<i>Rishal Ravikesh Chand (The University of Fiji, Fiji), ABM Shawkat Ali (The University of Fiji, Fiji), and Peng Zhang (Shenzhen University, China)</i>	
Group-Oriented Multi-signature Supporting Monotonic Endorse Policies in Hyperledger Fabric....	256
<i>Peng Zhang (Shenzhen University, China), Yongwen Huang (Shenzhen University, China), Fa Ge (Shenzhen University, China), and Yuhong Liu (Santa Clara University, USA)</i>	

Blockchain-10: Applications and Services (III)

A Method and Platform for Security Advisory Dissemination Leveraging Web3 Technologies	265
<i>Jannik Lucas Sommer (Aalborg University, Denmark), Magnus Mølgaard Lund (Aalborg University, Denmark), Nicola Cibir (Aalborg University, Denmark), and Michele Albano (Aalborg University, Denmark)</i>	
PureLottery: Fair Leader Election Without Decentralized Random Number Generation	273
<i>Jonas Ballweg (Technical University of Munich, Germany), Zhuo Cai (Hong Kong University of Science and Technology, China), and Amir Goharshady (Hong Kong University of Science and Technology, China)</i>	
Smart Parking Dapp: A Trust Minimised Smart Parking Solution	281
<i>Nazim Rizvic (Monash University), Amin Sakzad (Monash University), Arash Mirzaei (Monash University), and Taha Hossein Rashidi (The University of New South Wales)</i>	

Efficiency-Enhanced Blockchain-Based Client Selection in Heterogeneous Federated Learning	289
<i>Zhiqi Lei (School of Computer Science and Technology, Beijing Institute of Technology, China; Yangtze Delta Region Academy of Beijing Institute of Technology, China), Keke Gai (School of Cyberspace Science and Technology, Beijing Institute of Technology, China; Yangtze Delta Region Academy of Beijing Institute of Technology, China), Jing Yu (Institute of Information Engineering, Chinese Academy of Sciences, China), Shuo Wang (School of Cyberspace Science and Technology, Beijing Institute of Technology, China; Yangtze Delta Region Academy of Beijing Institute of Technology, China), Liehuang Zhu (The School of Cyberspace Science and Technology, Beijing Institute of Technology, China; Yangtze Delta Region Academy of Beijing Institute of Technology, China), and Kim-Kwang Raymond Choo (Department of Information Systems and Cyber Security, University of Texas at San Antonio, USA)</i>	
BBB-Voting: Self-Tallying End-to-End Verifiable 1-out-of-k Blockchain-Based Boardroom Voting	297
<i>Ivan Homoliak (Brno University of Technology), Zengpeng Li (Shandong University), and Pawel Szalachowski (Singapore University of Technology and Design)</i>	

Blockchain-11: New Architectures

Decision Models for Selecting Patterns in Governance-Driven Blockchain Systems	307
<i>Yue Liu (Data61, CSIRO, Australia; University of New South Wales, Australia), Qinghua Lu (Data61, CSIRO, Australia; University of New South Wales, Australia), Hye-Young Paik (University of New South Wales, Australia), Guangsheng Yu (Data61, CSIRO, Australia), and Liming Zhu (Data61, CSIRO, Australia; University of New South Wales, Australia)</i>	
Securing Smart UAV Delivery Systems Using Zero Trust Principle-Driven Blockchain Architecture	315
<i>Chengzu Dong (Deakin University, Australia), Shantanu Pal (Deakin University, Australia), Qi An (Deakin University, Australia), Aiting Yao (Anhui University, China), Frank Jiang (Deakin University, Australia), Zhiyu Xu (Swinburne University of Technology, Australia), Jianhua Li (Deakin University, Australia), Meiqu Lu (Guangxi Minzu University, China), Yangxu Song (Deakin University, Australia), Shiping Chen (CSIRO, Data61, Australia), and Xiao Liu (Deakin University, Australia)</i>	
Account Abstraction, Analysed	323
<i>Qin Wang (CSIRO Data61, Australia) and Shiping Chen (CSIRO Data61, Australia)</i>	
Crisis of Trust: Analyzing the Verifier's Dilemma in Ethereum's Proof-of-Stake Blockchain	332
<i>Daria Smuseva (Ca' Foscari University of Venice, Italy), Ivan Malakhov (Ca' Foscari University of Venice, Italy), Andrea Marin (Ca' Foscari University of Venice, Italy), and Sabina Rossi (Ca' Foscari University of Venice, Italy)</i>	

A Blockchain-Based Approach to Improving Smart Home Security with Situation-Aware Access Control	340
<i>Zhicheng Lin (Arizona State University, USA) and Stephen S. Yau (Arizona State University, USA)</i>	

Blockchain-12: Scalability and Optimization

Towards Scalable Cross-Chain Messaging	348
<i>João Otávio Chervinski (Monash University, Australia), Diego Kreutz (Monash University, Australia; Federal University of Pampa, Brazil), and Jiangshan Yu (Monash University, Australia)</i>	
SCU: A Hardware Accelerator for Smart Contract Execution	356
<i>Tao Lu (Louisiana State University, USA) and Lu Peng (Tulane University, USA)</i>	
Cross-Chain Based Spectrum Resources Transfer Technology Across Different Telecom Operators Towards 6G	365
<i>Mengjiang Liu (Beihang University, China), Qianhong Wu (Beihang University, China), Kun Wang (Beihang University, China), Tianxu Han (Beihang University, China), and Yiming Hei (Beihang University, China)</i>	
A Community-Based Strategy for Blockchain Sharding: Enabling More Budget-Friendly Transactions	370
<i>Zixu Zhang (University of Technology, Australia), Ying Wang (Hangzhou Dianzi University, China), Guangsheng Yu (Data61, CSIRO, Australia), Xu Wang (University of Technology, Australia), Ming Zhang (Xidian University, China), Wei Ni (Data61, CSIRO, Australia), and Renping Liu (University of Technology, China)</i>	
Fault-Tolerant Verifiable Encrypted Search on Blockchain	377
<i>Xuan Chen (Monash University, Australia) and Shujie Cui (Monash University, Australia)</i>	
Author Index	385